HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed, developed, and validated predictive models using **Python** and **scikit-learn**, achieving an average **AUC of 0.88** on held-out customer churn data.
- Transformed complex, large-scale datasets from relational databases using advanced **SQL**, building scalable data structures for downstream machine learning model training.
- Collaborated with product owners and stakeholders in planning sessions to define business challenges and recommend appropriate **data science modeling** approaches, translating ambiguous requirements into actionable plans.
- Managed priorities across multiple concurrent projects, shaping tradeoffs and cross-functional team efforts to deliver on critical go-to-market intelligence initiatives.
- Contributed to the development and validation of machine learning products by working closely with the machine learning engineering team, ensuring alignment with technical design and deployment strategies.
- Developed technical design documentation for data science and AI products, detailing architecture, methodology, and validation criteria for stakeholder review and approval.
- Containerized and deployed data science applications using **Docker** and managed cloud resources on **AWS EC2**, ensuring scalable and reliable execution of analytical workloads.

SKILLS

Machine Learning – model development, validation, predictive modeling, scikit-learn, statistical methods, applied AI

Programming & Data Manipulation – Python, SQL, pandas, NumPy, R

Data Warehousing & ETL – large-scale data transformation, data pipelines, data modelling, Power Query

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD, Jenkins

BI & Visualization – Power BI, Tableau, DAX, data visualization, dashboards

Software Engineering – Java, Spring Boot, REST APIs, Git, Jira, Confluence

PROJECTS

Customer Segmentation Model for Go-to-Market Strategy

- Developed a customer segmentation model using **K-Means clustering** in **Python**, analyzing transaction history and user engagement data for approximately 150,000 customer records.
- Engineered feature sets including recency, frequency, and monetary value (RFM) scores, creating reusable data products that improved marketing campaign targeting accuracy by **18%**.
- Presented model findings and segmentation insights to the marketing stakeholder team, enabling data-informed strategic adjustments to customer outreach programs.

Predictive Lead Scoring for Sales Pipeline Enhancement

- Built and evaluated a **logistic regression model** using **scikit-learn** to predict lead conversion probability from a dataset of 50,000 historical sales leads.
- Optimized feature engineering by incorporating engagement signals and firmographic data, improving model **precision to 0.72** and recall to 0.65.
- Deployed the model's scoring logic via a **FastAPI** service, integrating with the existing CRM system for real-time lead prioritization.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)